

Designed for your needs

FreeCAD is designed to fit a wide range of uses including product design, mechanical engineering and architecture. Whether you are a hobbyist, a programmer, an experienced CAD user, a student or a teacher, you will feel right at home with FreeCAD.

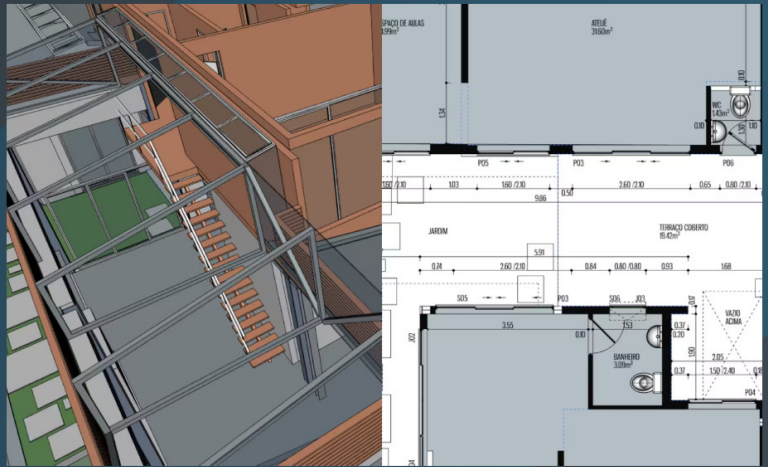


Figure 3: Building floorplans using FreeCAD

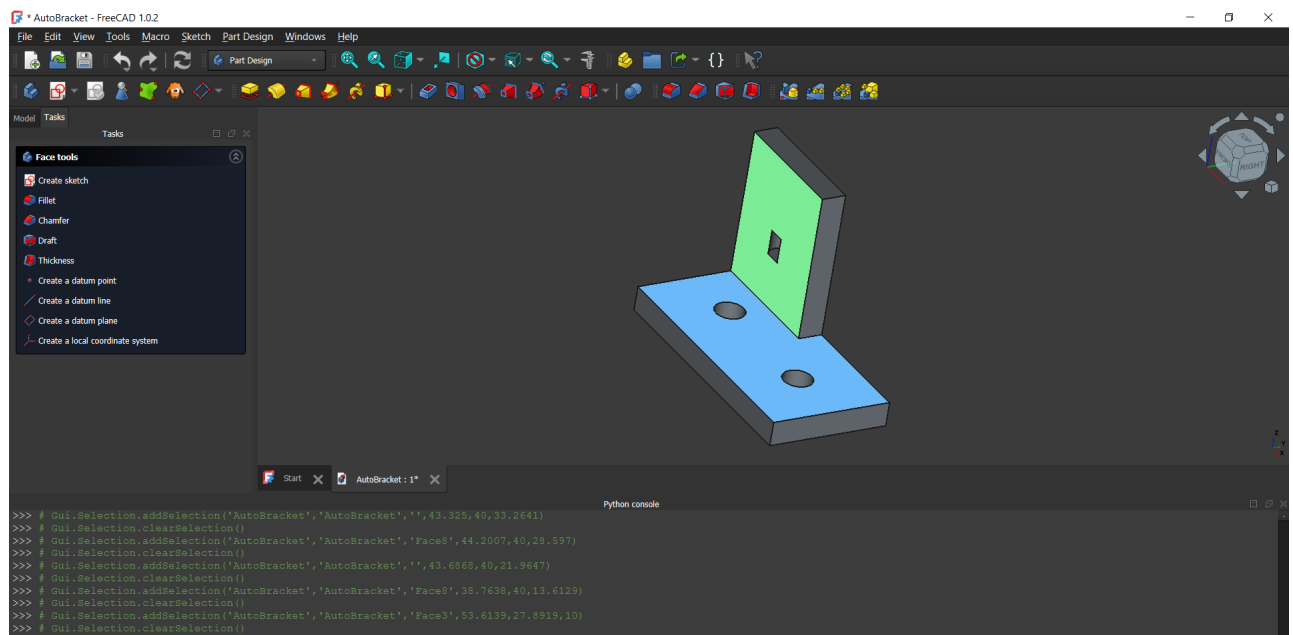


Figure 4: Automobile mounting bracket using FreeCAD and Python

Its key components are:

- Geometry engine
 - Solid modelling tools
 - Parametric modelling tools
 - Surface modelling tools
 - Rendering and visualisation
 - Constraints and relations
 - Assembly modelling
 - Simulation modules
 - Version control and collaboration
 - Drafting and annotation
 - File interoperability (STEP, STL, IGES, and many others)
- FreeCAD can be used to develop professional and

high-end structural designs and architectures. 2D and 3D geometries can be created, and details of the designs can be extracted.

FreeCAD integrates Python programming for designing and viewing 3D objects. Here's sample code of how Python and FreeCAD can be used for designing an automobile mounting bracket.

```
import FreeCAD
import Part

# Create document
mydoc = FreeCAD.newDocument("AutoBracket")
```